

DATA ANALYTICS CAPSTONE

Healthcare Infrastructure Gap Index (HIGI)

An AI-Assisted Decision Support Framework for District-Level Healthcare Infrastructure Prioritization in India

Anirudh Harathi

Mentor: Mr. Kartik Umesh Sharma

Executive Summary

OBJ

Objective

Develop a district-level, data-driven framework to measure healthcare infrastructure gaps and support prioritization across India.

DATA

Data

Integrated RHS, Census of India 2011, and NFHS-5 data covering 488 districts and 18 variables.

APP

Approach

Constructed the Healthcare Infrastructure Gap Index (HIGI), profiled districts with K-Means, and built an AI-assisted Random Forest prioritization model.

RES

Key Results

HIGI ranged 0.0861–0.9765; Random Forest achieved $R^2 = 0.7092$, MAE = 0.0524; three district profiles identified with moderate cluster separation.

USE

Usage

HIGI, district profiles, and AI-assisted rankings provide complementary evidence for planning, while retaining expert and policy judgement.

The framework converts fragmented district-level data into multiple evidence-based views for healthcare infrastructure prioritization.

Gap Analysis



IDENTIFIED GAP

Need for an integrated, district-level analytical framework that combines infrastructure measurement with statistical analysis and machine learning to support prioritization.

STUDY CONTRIBUTION

HIGI + District Profiling + AI-Assisted Ranking

The study combines these three complementary perspectives to inform decisions — rather than replacing policy judgement.

Research Questions

RQ	Research Question	Method	Key Result
RQ1	Can official district-level datasets be integrated to quantify infrastructure gaps?	Data integration & quality assessment	488 districts × 18 variables
RQ2	Can HIGI measure and compare relative infrastructure gaps?	HIGI construction & Spearman correlation	HIGI: 0.0861–0.9765
RQ3	Can ML identify meaningful district patterns?	K-Means clustering & validation	3 district profiles
RQ4	Can AI-assisted prioritization provide a complementary perspective?	Random Forest + OOF predictions + ranking comparison	$\rho = 0.653$; Top-20 overlap = 10%

Four analytical questions → measurement → district profiling → AI-assisted prioritization

Data Description & EDA

RHS

5 healthcare infrastructure indicators

Census 2011

4 demographic indicators

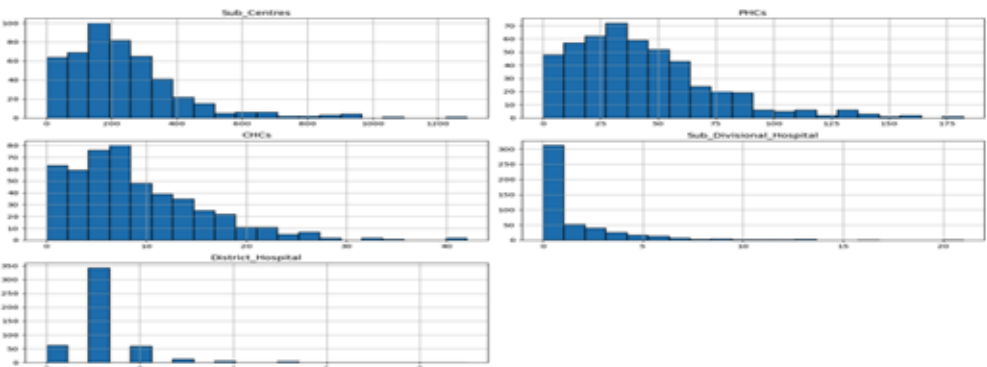
NFHS-5

7 health indicators

488 × 18

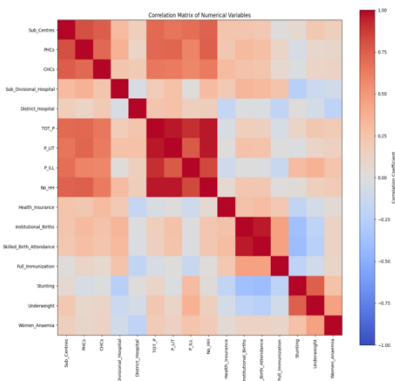
Districts × Variables

Distribution of Healthcare Infrastructure Variables



Infrastructure availability varies substantially across districts, with positively skewed distributions across facility types.

Correlation Matrix



Shading reflects relative association strength. Infrastructure variables show strong internal relationships and meaningful associations with demographic and health indicators. *HIGI ↔ Health Insurance: $p = -0.348$ (moderate negative association).*

DATA QUALITY

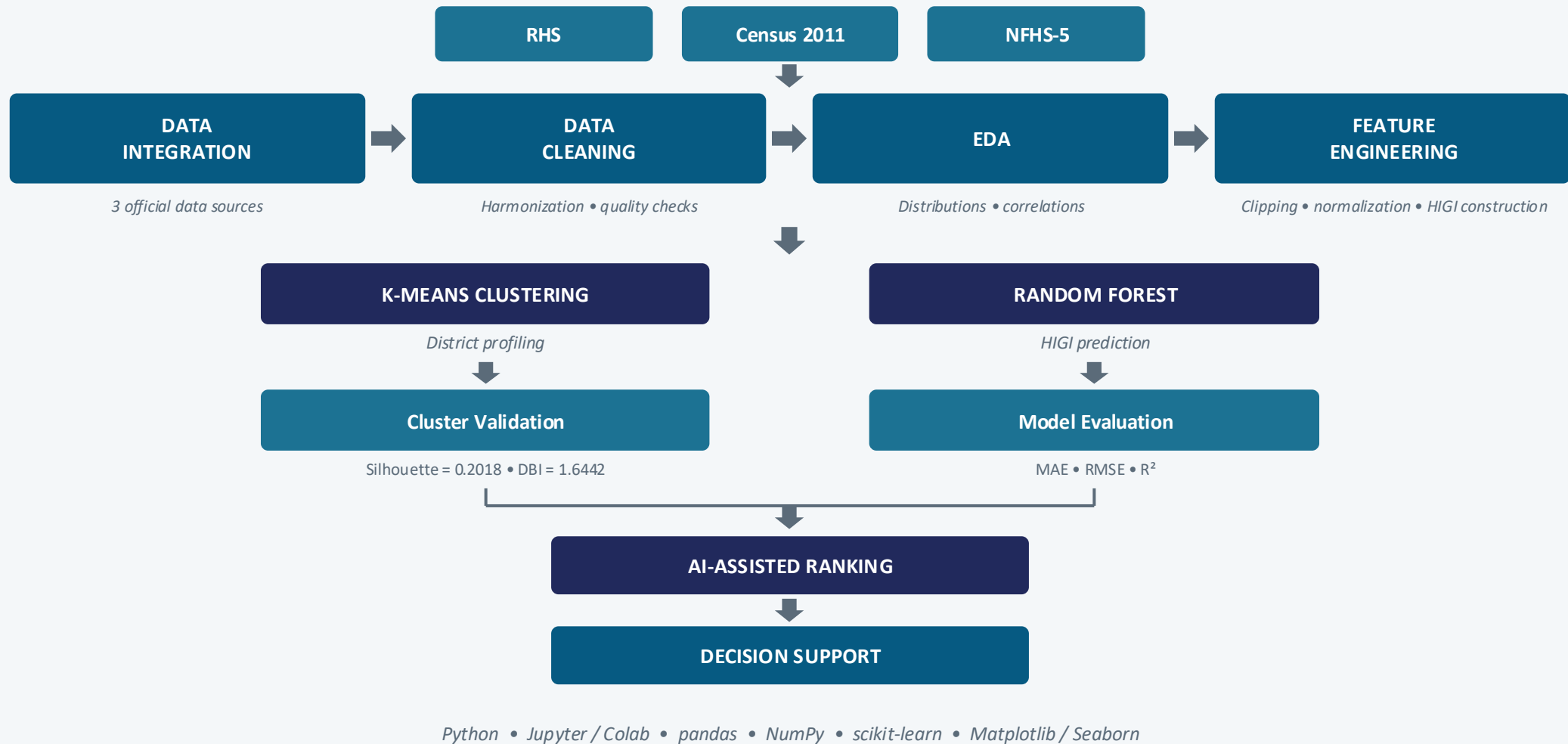
11 missing observations

Full Immunization: 9 · Stunting: 1 · Underweight: 1

No missing infrastructure or demographic values

KEY TAKEAWAY: Substantial variation in healthcare infrastructure, meaningful cross-variable relationships, and limited missingness support standardized district-level comparison and downstream modelling.

Architecture diagram/Workflow



Model Building

1 · HIGI Construction

Five infrastructure indicators — 20% each:

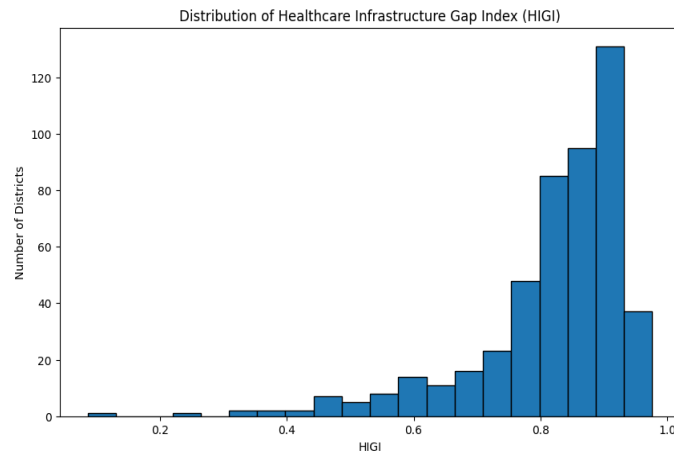
- Sub-Centres / 100k
- PHCs / 100k
- CHCs / 100k
- Sub-Divisional Hospitals / 100k
- District Hospitals / 100k

1st/99th pct. clipping → Min-Max normalization → Direction reversal → Composite HIGI

Higher HIGI = larger relative infrastructure gap

Why HIGI?

Transparent • Comparable • Population-normalized



2 · District Profiling

K-Means Clustering

- K evaluated: 2–8 → selected K = 3
- StandardScaler applied
- PCA used only for visualization

Validation

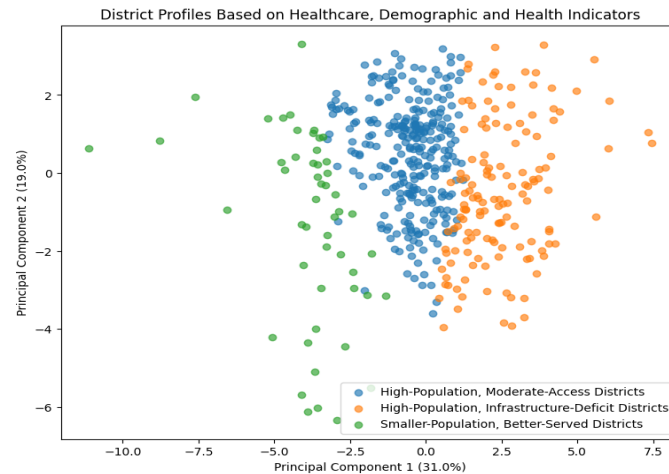
Silhouette = 0.2018

Davies–Bouldin = 1.6442

3 interpretable district profiles identified; cluster separation was moderate.

Why K-Means?

Unsupervised • Numerical data • Interpretable profiles



3 · Predictive Model

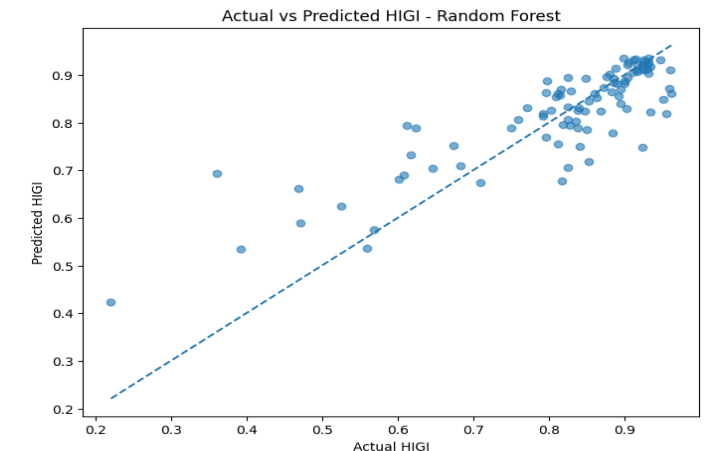
Random Forest Regression

- Captures non-linear relationships
- Handles feature interactions
- More robust than a single decision tree
- Appropriate for structured/tabular data

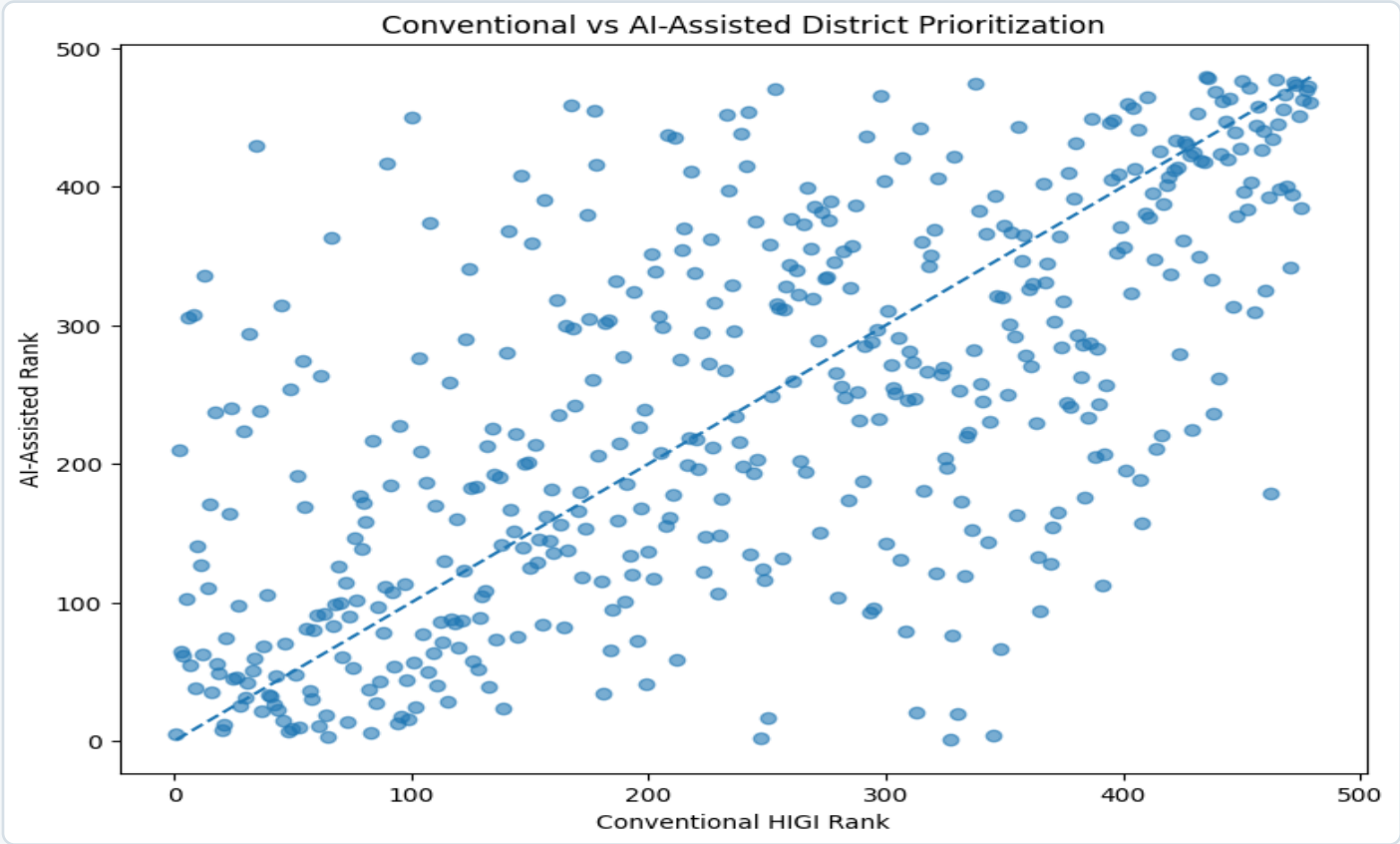
Leakage control: HIGI's five construction indicators excluded from RF predictors.

Why Random Forest?

Nonlinear relationships • Interaction-aware • Robust for tabular data



Results



RANDOM FOREST PERFORMANCE

Model	MAE↓	RMSE↓	R ² ↑
Baseline	0.0990	0.1429	-0.0013
Random Forest	0.0524	0.0770	0.7092
Tuned RF	0.0529	0.0773	0.7073

$\rho = 0.653$

Spearman Correlation | Moderate rank agreement

10%

Top-20 overlap

2 / 20

Common top-20 districts

AI-assisted ranking provides a complementary prioritization signal: not a replacement for HIGI.

Implementation and User Benefit

- 1 Identify** Use HIGI rankings to identify districts with relatively larger infrastructure gaps.
- 2 Profile** Use K-Means district profiles to understand differences in population, infrastructure, and health characteristics.
- 3 Prioritize** Use AI-assisted Random Forest rankings as an additional signal for districts requiring further review.
- 4 Integrate** Combine analytical outputs with budget, feasibility, local context, and policy priorities.
- 5 Decide** Provide evidence to policymakers and domain experts while retaining human judgement in final allocation decisions.

EXAMPLE USE CASE

A state health department could screen districts using HIGI, segment them by district profile, and use the AI-assisted ranking to flag districts for deeper feasibility assessment before allocating infrastructure resources.

One District → Multiple Evidence-Based Perspectives → Better-Informed Review

Conclusion

01

Integration

Successfully integrated 488 districts × 18 variables from RHS, Census 2011, and NFHS-5.

02

Measurement

HIGI provides a standardized measure of relative healthcare infrastructure gaps across districts.

03

Segmentation

K-Means identified 3 district profiles, providing contextual segmentation.

04

AI-Assisted Prioritization

Random Forest achieved $R^2 = 0.7092$; AI-assisted ranking showed $p = 0.653$ with only 10% Top-20 overlap.

LIMITATIONS & FUTURE SCOPE

- Data are district-level and from different reference periods
- Framework excludes real-time and private healthcare infrastructure
- Cluster separation is moderate; model shows association, not causation
- Future work: newer datasets, HIGI weight sensitivity analysis, feasibility variables, refreshed-data deployment

HIGI should be viewed as a decision-support framework, not an automated allocation system: combining infrastructure gaps, district profiles, and AI-assisted evidence to support expert decision-making.

Bibliography

- International Institute for Population Sciences (IIPS), & ICF. (2021). National family health survey (NFHS-5), 2019–21: India. IIPS.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An introduction to statistical learning: With applications in R (2nd ed.). Springer.
- Ministry of Health and Family Welfare. (2022). Rural health statistics 2021–22. Government of India.
- Mukherji, A., Rao, M., Desai, S., Subramanian, S. V., Kang, G., & Patel, V. (2024). District-level monitoring of universal health coverage, India. *Bulletin of the World Health Organization*, 102(9), 630–638B.
- Office of the Registrar General & Census Commissioner, India. (2011). Census of India 2011. Government of India.
- Organisation for Economic Co-operation and Development, & Joint Research Centre. (2008). Handbook on constructing composite indicators: Methodology and user guide. OECD Publishing.
- Russell, S., & Norvig, P. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson.
- Sehgal, M., Jatrana, S., & Johnson, L. (2024). A comprehensive health index for India: Development, validation, and spatial variation. *Journal of Population Research*, 41, Article 21.
- Sharda, R., Delen, D., & Turban, E. (2023). Business intelligence, analytics, data science, and AI: A managerial perspective (5th ed.). Pearson.
- Sharma, A., Prinja, S., & Aggarwal, A. K. (2017). Measurement of health system performance at district level: A study protocol. *Journal of Public Health Research*, 6(3), Article 917. <https://doi.org/10.4081/jphr.2017.917>
- Sharma, A., Prinja, S., & Aggarwal, A. K. (2019). Comprehensive measurement of health system performance at district level in India: Generation of a composite index. *The International Journal of Health Planning and Management*, 34(4), e1783–e1799.
- Sivasubramanian, S., & Raval, A. (2025). Artificial intelligence–augmented public health interventions in India. *Health Affairs Scholar*, 3(5).
- Saragadam, S. K., Saragadam, S. H., Kumari, K. S. S., & Kasturi, G. V. K. (2025). Explainable machine learning on health management information system data to unveil health factors affecting maternal mortality ratio of districts in India towards achieving sustainable development goals. *Journal of Information Systems Engineering & Management*, 10(24s).